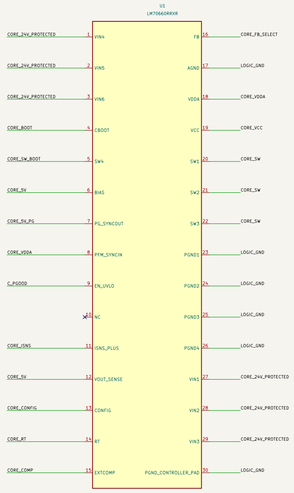
CORE 5 V / 6 A CONVERTER CANDIDATE — ENGINEERING CIRCUIT

LM70660: protected nominal24V input; no display/AL/drive power. C_PGOOD is analog enable, not GPIO.
Upstream R9 must be absent. VIN powers internal startup bias; no running host/ESP is needed to enable this converter.

























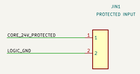


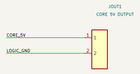










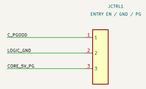


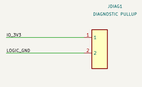
































Layout: Kelvin ISNS+/VOUT across RS1; separate quiet AGND returns join local power ground as TI specifies.
SW1 bootstrap return is internally common with SW1–3; preserve local bootstrap loop. No external long SW1 strap.
U1 RRK, RS1 long-side0612 and polarized bulk footprints unassigned; no generic substitutions.
Bulk = three MLCCs require delayed SBC branch and revised compensation. Transient margin is conditional.
Compensation, negative clamps and reverse feed remain unqualified. No finished PCB or machine permission.

Sheet: /	
File: core_converter.kicad_sch	
Title: Core 5V converter candidate	
Size: A1	Date: 2026-09-28
KiCad E.D.A. 10.0.4	**Rev: r0.1**
	Id: 1/1